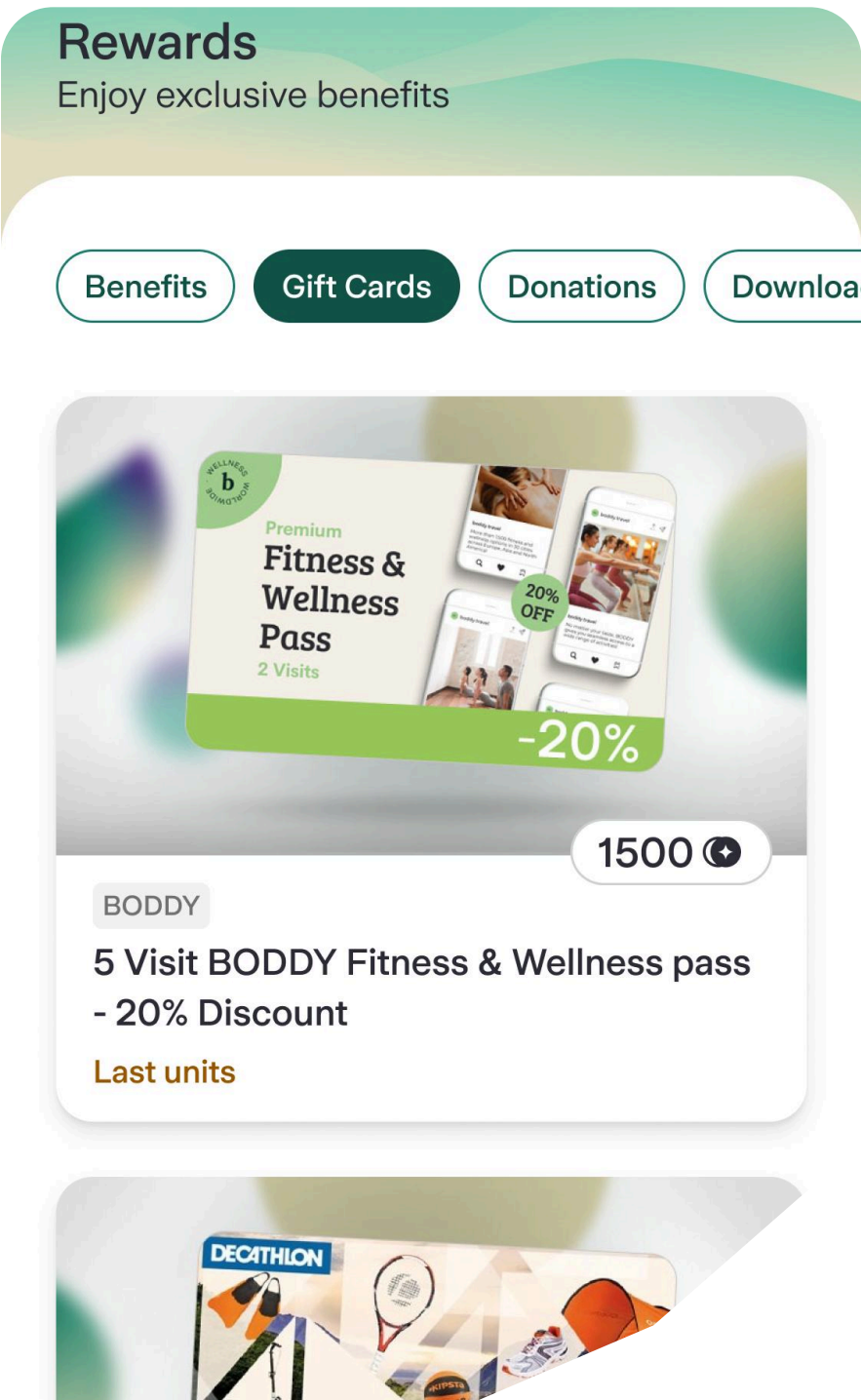
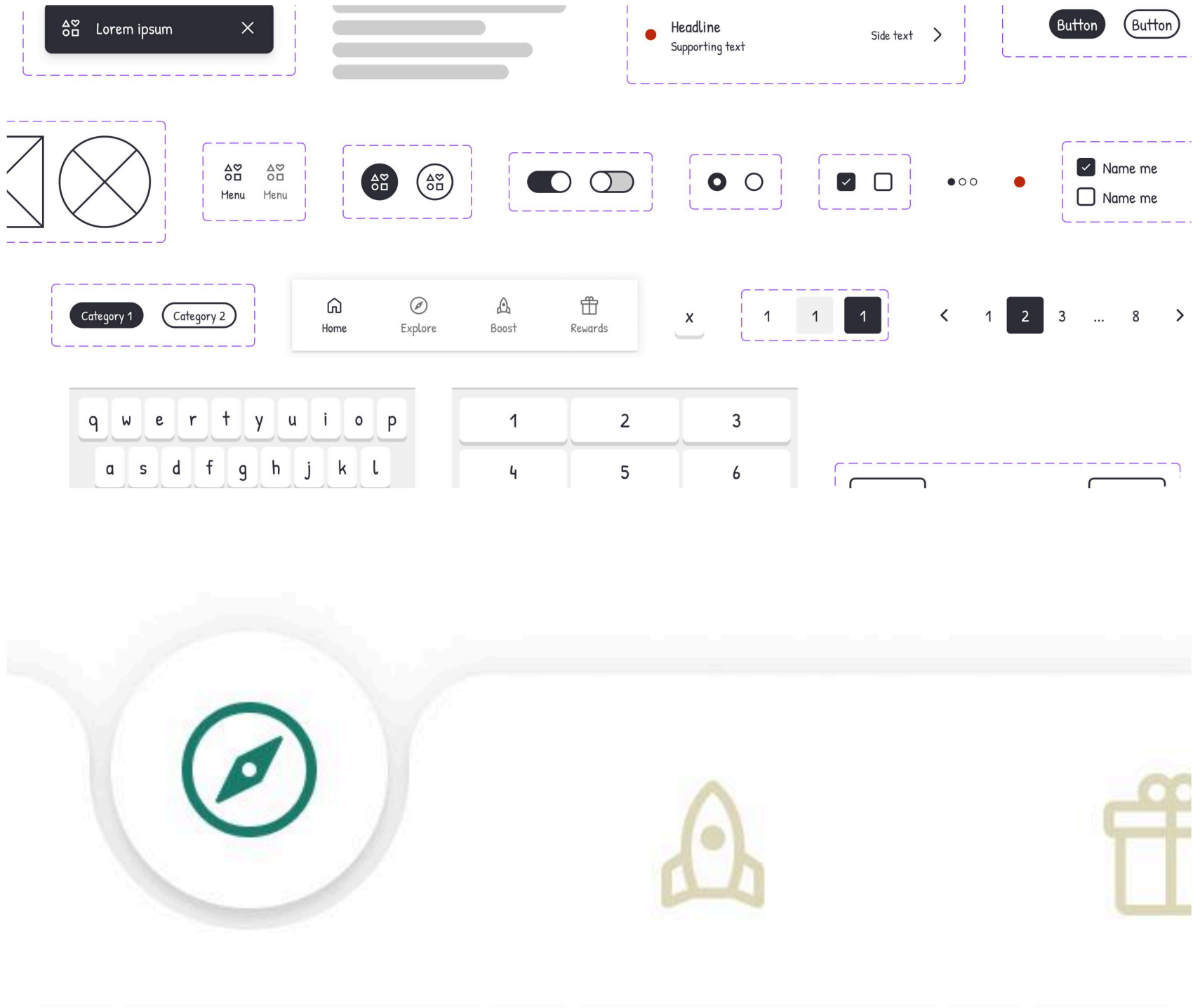


October 2025



Design System

# Behind the scenes

By Anso & Jeni

What is a Design System?





# Design System = Design + Code

It's not just about how things look, but also how they behave, how they're built, and how they scale.

- Foundations
- Components
- Patterns
- Guidelines
- Code
- Governance

Design sees a system of rules, decisions and visuals.

Dev sees a system of reusable code, theming & cross-platform consistency.

Together it's a single source of truth that scales across apps.

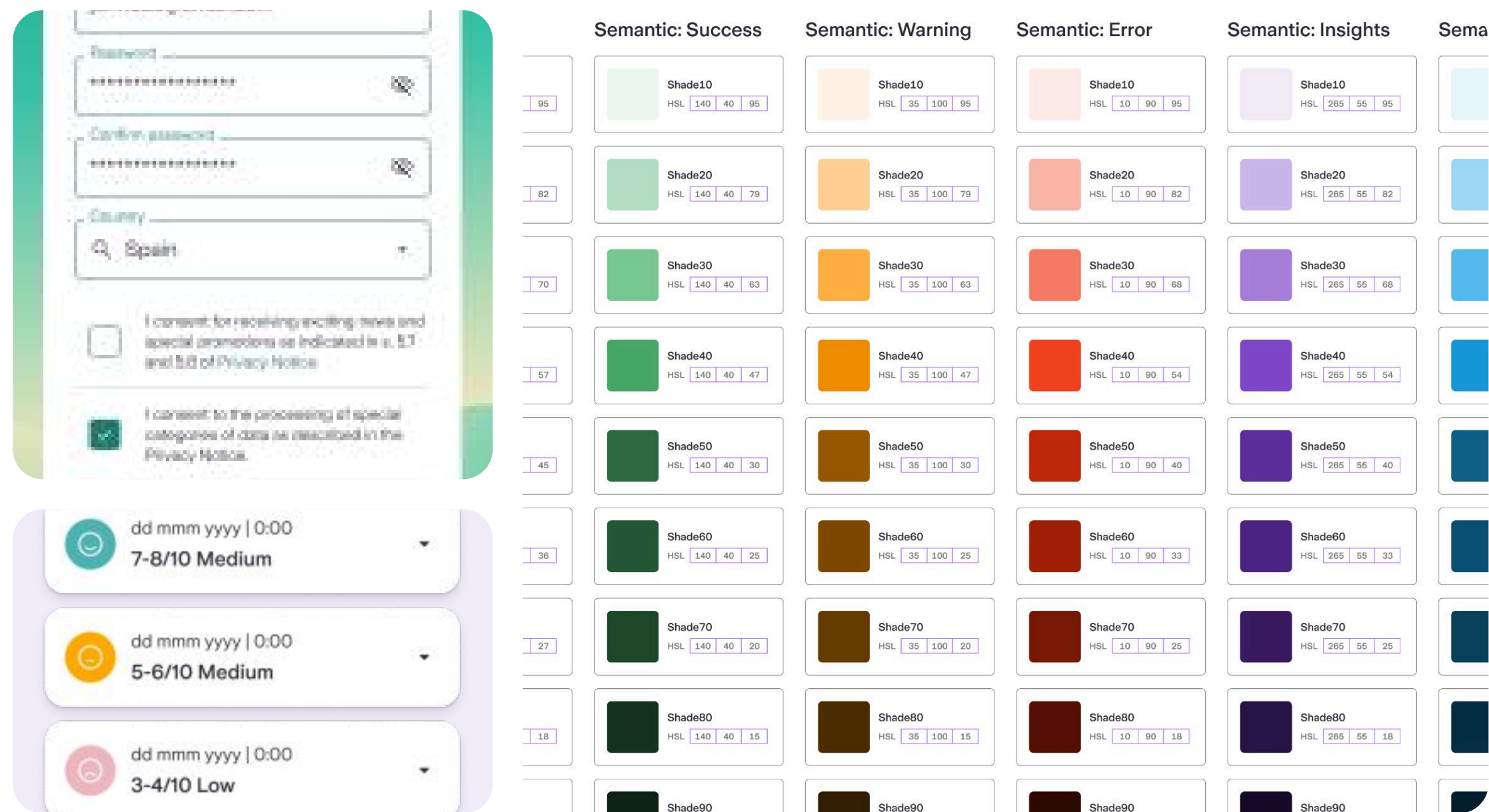




# Design Tokens & Theme

# Design Tokens: The DNA of Design System

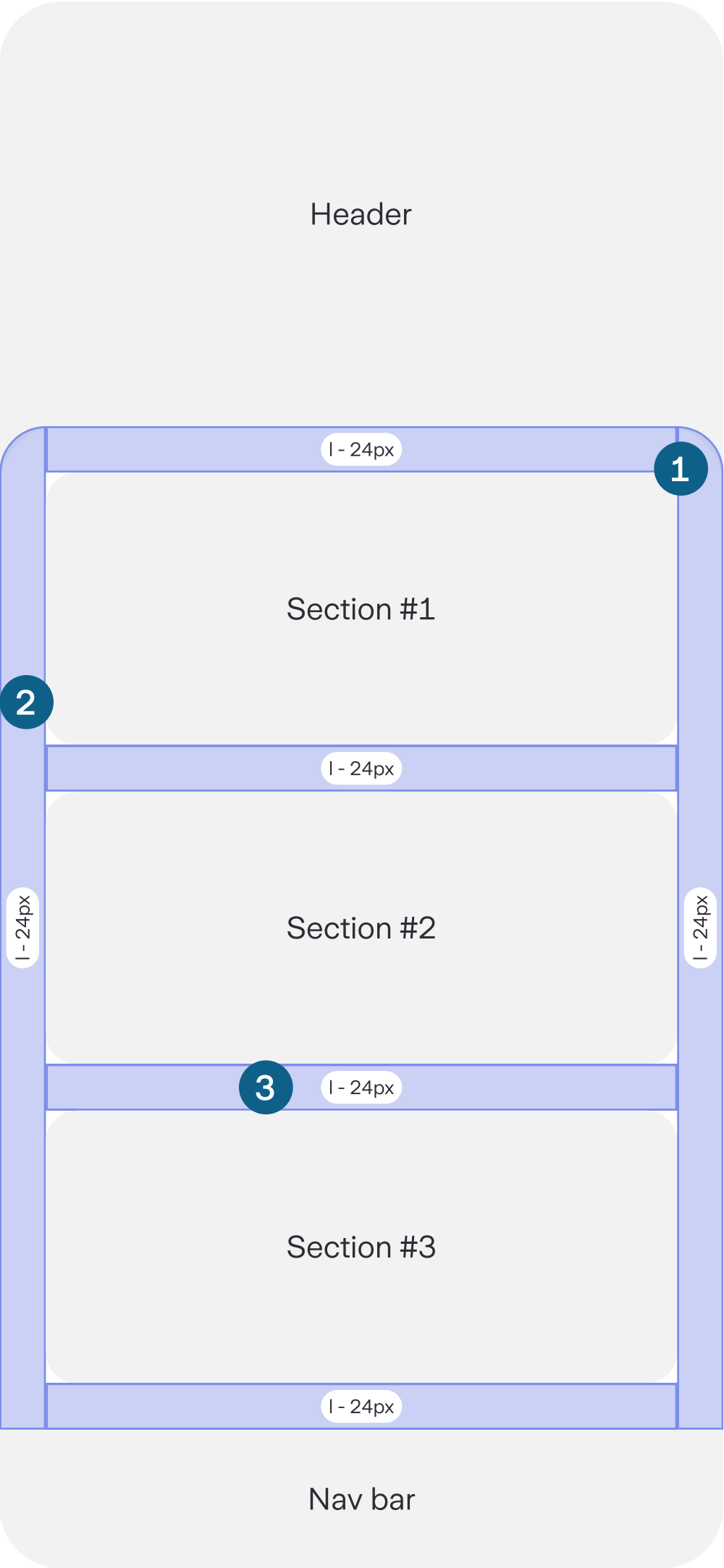
Change one token, update entire ecosystem instantly



- **Colors:** (row of swatches) Primary, secondary, semantic (success, error, warning..)
- **Typography:** Font families, sizes, weights, line heights
- **Units:** Spacing, sizes & radius

```
DesignSystemTheme.of(context).colors
├── primary      → buttons, accents
├── background   → app background
├── surface      → elevated containers
├── onSurface    → text on background/surface
├── onPrimary    → text/icons on primary backgrounds
└── onColor     → text/icons on colored backgrounds
```

# Design rules



## Icons

|    |            |
|----|------------|
| xs | 16 x 16 px |
| s  | 20 x 20 px |
| m  | 24 x 24 px |
| l  | 32 x 32 px |
| xl | 40 x 40 px |

## Radius

|      |         |
|------|---------|
| xxs  | 2 px    |
| xs   | 4 px    |
| s    | 8 px    |
| m    | 16 px   |
| l    | 24 px   |
| xl   | 32 px   |
| full | 1000 px |

## Spacing

|      |       |
|------|-------|
| xxxs | 2 px  |
| xxs  | 4 px  |
| xs   | 8 px  |
| s    | 12 px |
| m    | 16 px |
| mp   | 20 px |
| l    | 24 px |
| xl   | 32 px |
| xxl  | 40 px |
| xxxs | 48 px |

Container  
Section #1  
Section #2  
Section #3

- 1 The top border radius for the main content area (white background) is **size xl (32px)**
- 2 The padding all around (4 sides) the main content area (white background) is **size l (24px)**
- 3 The distance between sections is **size l (24px)**

Personal Information

John Doe  
Joined Sept 2022

Account information  
Email: antoniosmith@gmail.com  
Country: Spain

General health check  
How healthy are you?

Personal information  
Date of birth: E.g. 15 October 1981  
Weight (kg): E.g. 75  
Height (cm): E.g. 170  
Do you smoke? Yes No  
Save profile  
Delete account

1

2

4

5

3

Personal Information

John Doe  
Joined Sept 2022

Account information  
Email: antoniosmith@gmail.com  
Country: Spain

General health check  
How healthy are you?

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Date of birth: E.g. 15 October 1981  
Weight (kg): E.g. 75  
Height (cm): E.g. 170  
Do you smoke? Yes No  
Save profile  
Delete account

1

2

4

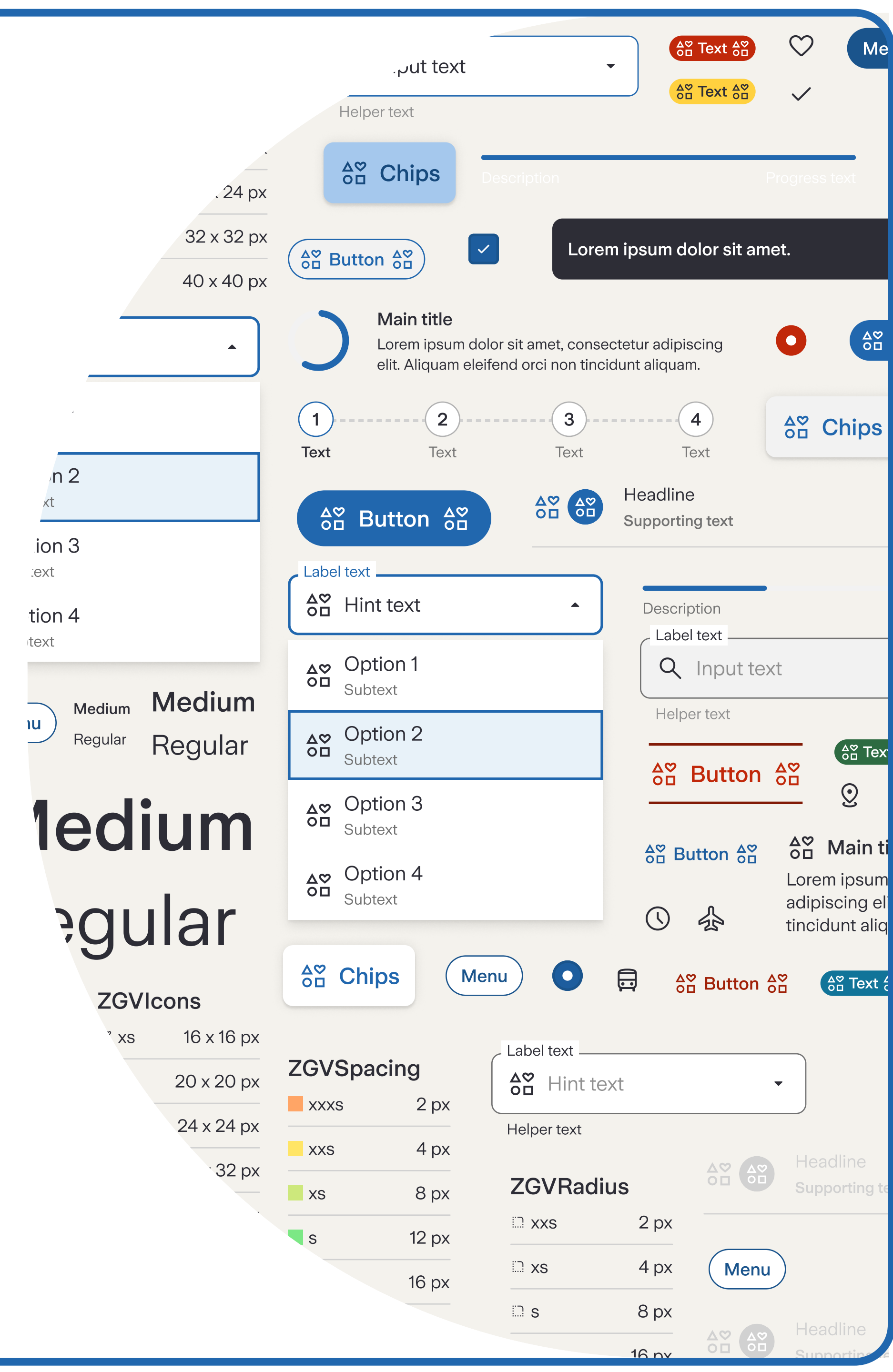
5

3



# Components - The Building Blocks

- Reusable UI elements (buttons, inputs, cards, etc.)
- Designed once, implemented once - Used everywhere
- Built through composition - Small parts form larger patterns
- Consistent look & behavior, across all apps and platforms
- Accessible & tested out of the box
- Accelerators - Speed up development (focus on feature, not UI details)

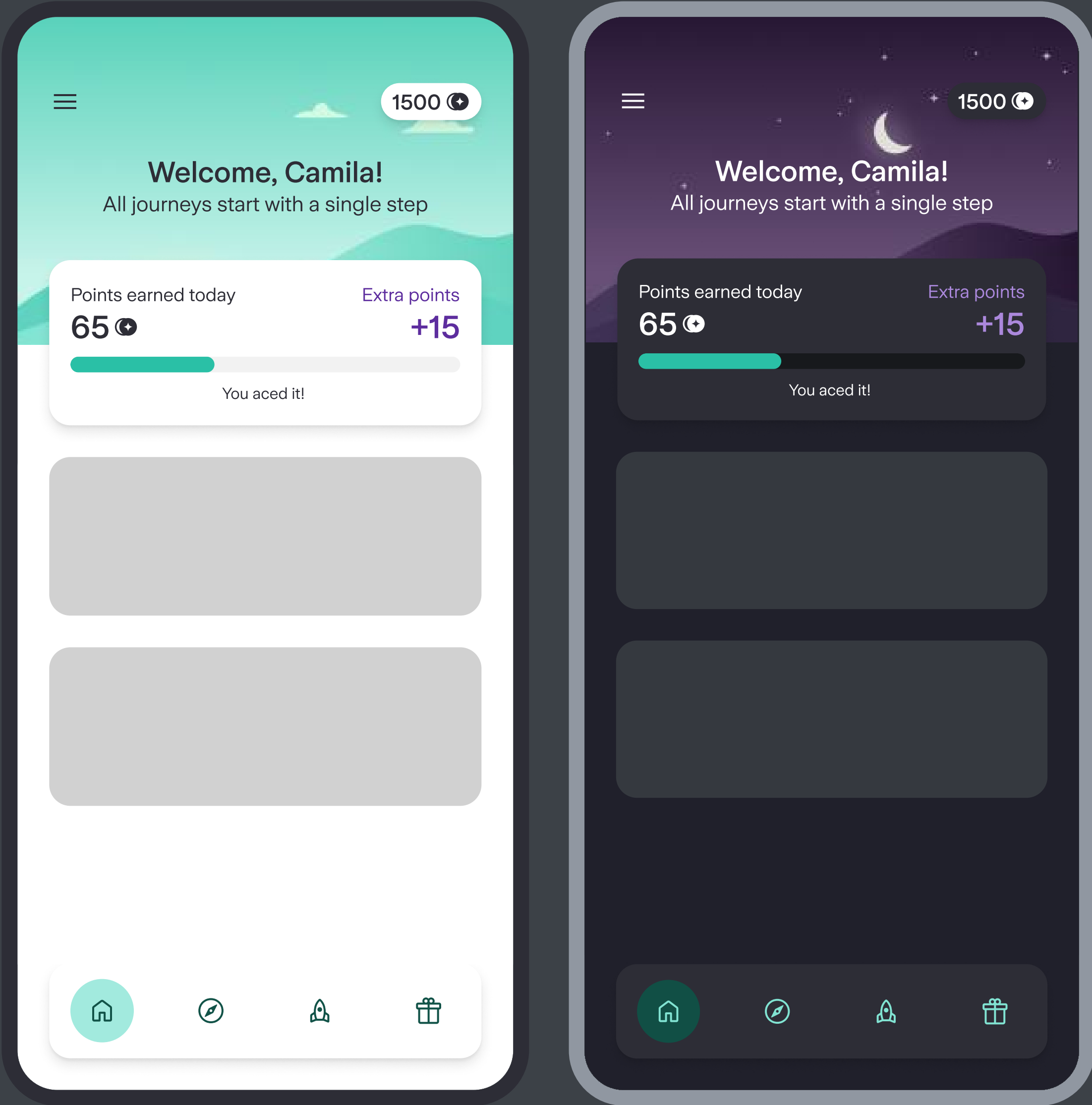


DESIGN TOKENS

# Dark Mode

By default, not as an afterthought.

One extendable Design SystemTheme supporting both dark and light mode.





# Development Process

## Dedicated Repository

- zgv\_design\_system repo (Flutter plugin)
- Contains both the **library** code and an **example** app

## Example App

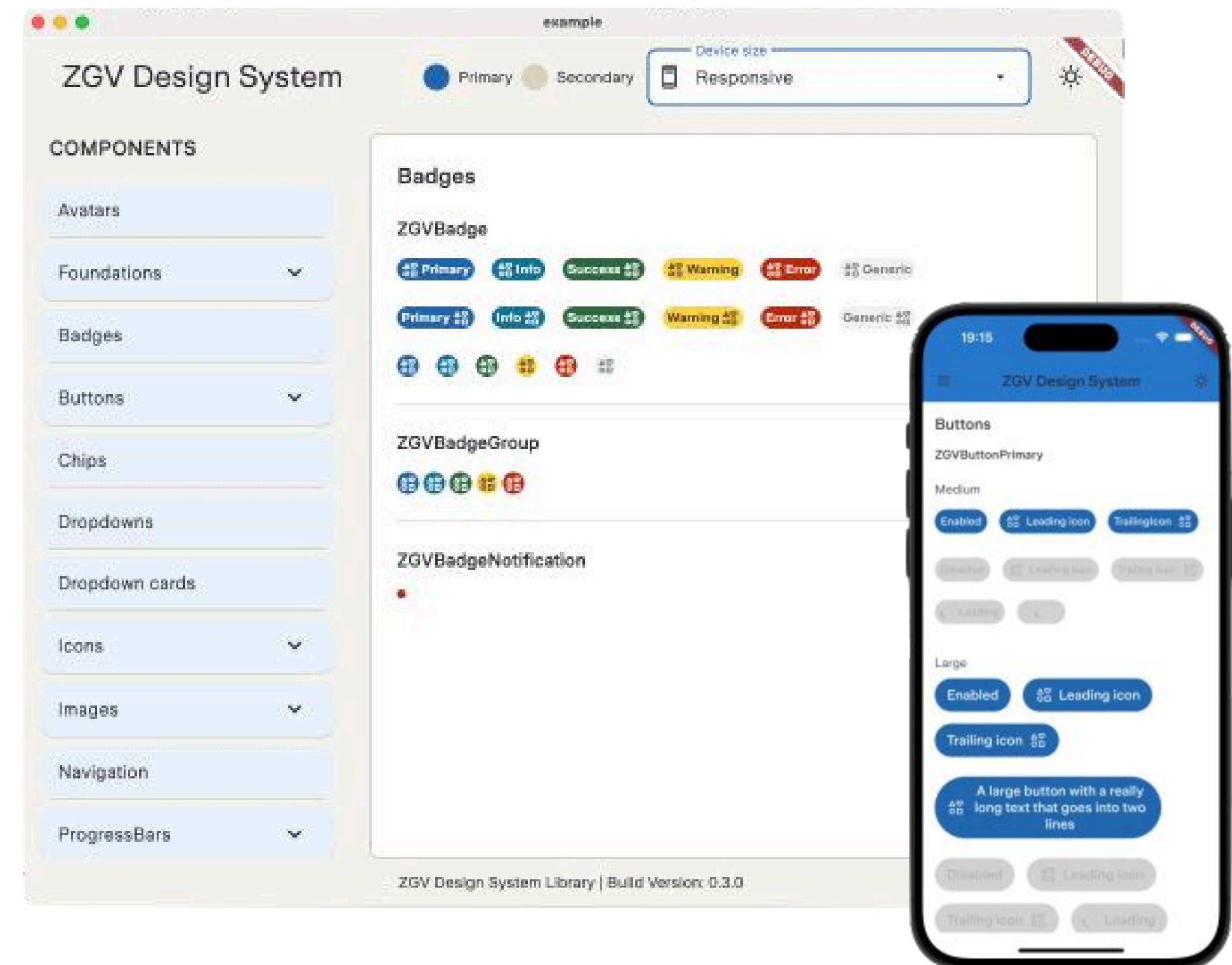
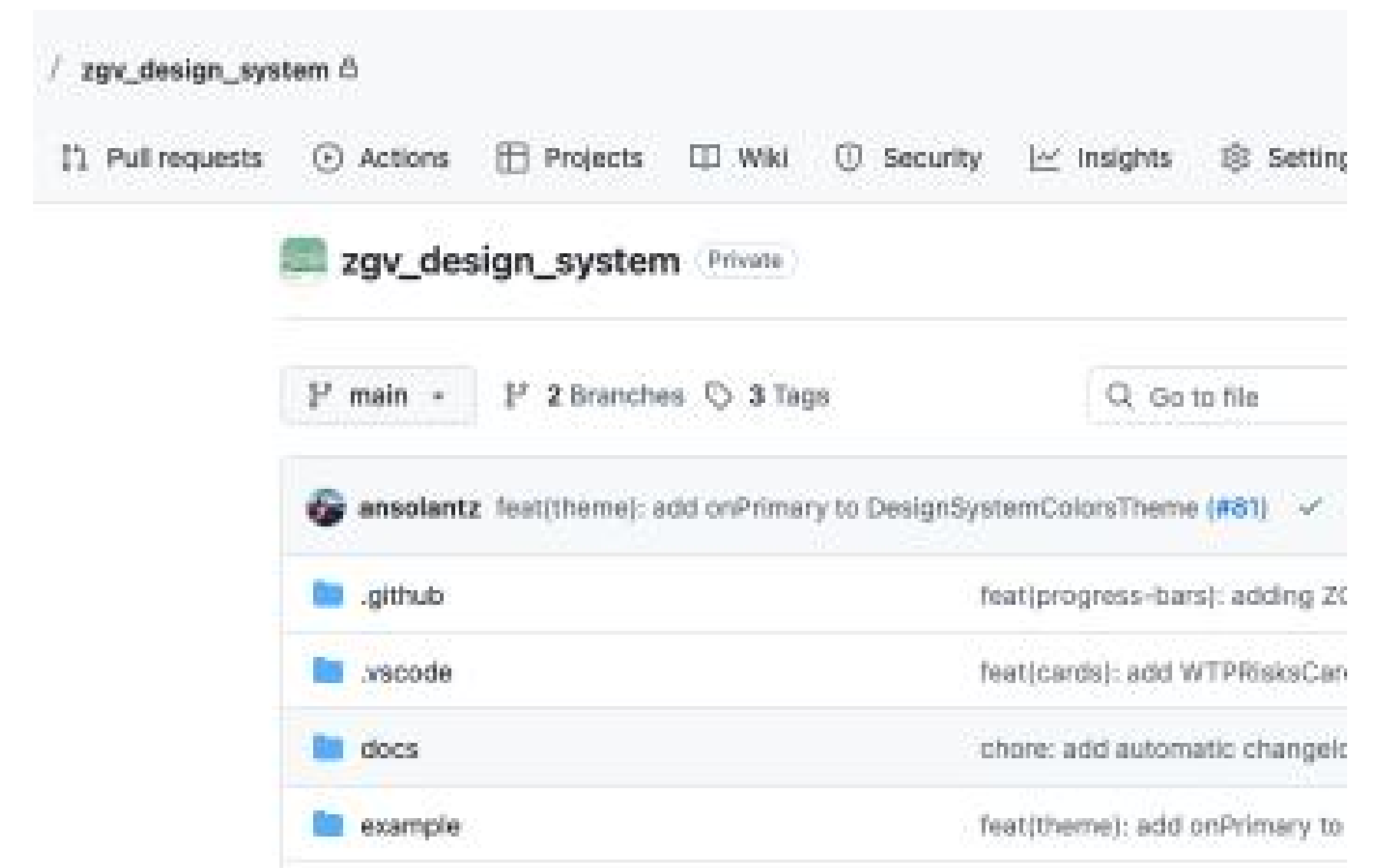
- Showcases **all components** in one place
- Used for manual testing (visuals + accessibility)

## Golden Tests

- Each component demo has a golden test
- Detects any visual change or regression automatically

## Result

- Reliable, tested components before adoption in products



UI optimisation and benefits for users



Reducing cognitive load for a user is essential in delivering a seamless experience

Increasing the amount of different components means higher cognitive effort from the users to understand how to engage with the app

**Personal Information**

Name  
Antonia Smith

Email  
antoniasmith@gmail.com

Biological sex  
Female

Gender identity  
Female

Date of birth  
11 October 1981

Weight (kg)  
68kg

Height (cm)  
167cm

How often do you drink alcohol?  
Never

Do you smoke?  
☐ Yes ☒ No

[Delete account](#)

**My Groups**

You are part of the following groups:

- LiveWell Users - India
- Polar - Spain

By being part of this group, you receive additional benefits such as extra rewards, content, and boosts.

[Remove group](#)

[Have a code to join another group?](#)

**Contact Support**

Please select a topic most relevant to your issue below and share any specific details in the next column. If applicable, please upload any screenshots of your issue.

How can we help you?

Share details

[Attach a file](#)

[Submit](#)

A Design System holds many benefits once implemented successfully:

- **Consistency** across app(s) for all components and branding
- Safe guard **standards** such as **accessibility**
- **47% faster deployments and updates** across all assests using DS

The 'Personal Information' screen features a teal header with a back arrow. Below the header, a user profile card for 'John Doe' (Joined Sept 2022) is displayed. Underneath, an 'Account information' section shows the email 'antoniosmith@gmail.com' and country 'Spain'. A 'Personal information' section contains input fields for 'Date of birth' (with example 'E.g. 15 October 1981'), 'Weight (kg)' (with example 'E.g. 75'), and 'Height (cm)' (with example 'E.g. 170'). It also includes dropdown menus for 'Biological sex' (labeled 'Select biological sex') and 'Gender' (labeled 'Select gender'). At the bottom, a partially visible question 'How often do you drink alcohol?' is shown.

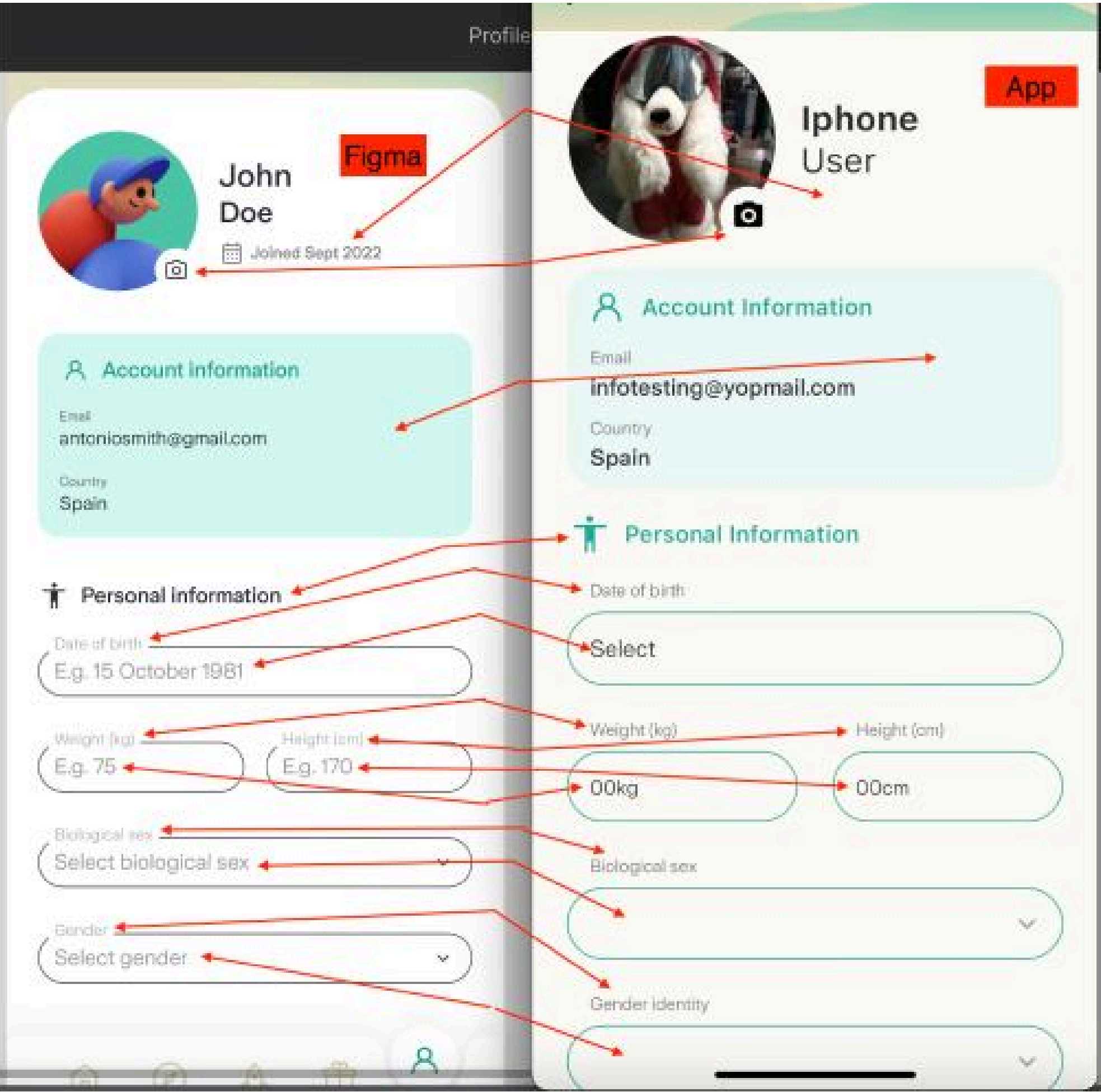
The 'My Groups' screen has a teal header with a back arrow. It displays a list of groups under the heading 'You are part of the following groups:'. The first group is 'LiveWell Users - Spain', which includes a description: 'LiveWell serves as the foundational group within your account, granting you access to all the basic functionalities of the LiveWell app. By being part of this group, you are one step closer to making health a habit. Join more groups to unlock additional benefits.' The second group is 'Polar - Spain'. The third group is 'Google', which includes a description: 'By being part of this group, you receive additional benefits such as extra rewards, content, and boosts.' and a 'Remove group' button. At the bottom, there is a green button labeled 'Add new group'.

The 'Contact Support' screen features a teal header with a back arrow. It prompts the user to 'Please select the topic most relevant to your issue below and share any specific details. If applicable, please upload any screenshots or a screen recording.' There is a dropdown menu for 'Issue' with the option 'How we can help you?'. Below this is a text area for 'Description' with the placeholder 'Please provide a detailed description of the issue.' and a character count '0/1000'. At the bottom, there is a button labeled 'Attach a file' and a large grey button labeled 'Send'.



Why We Need One

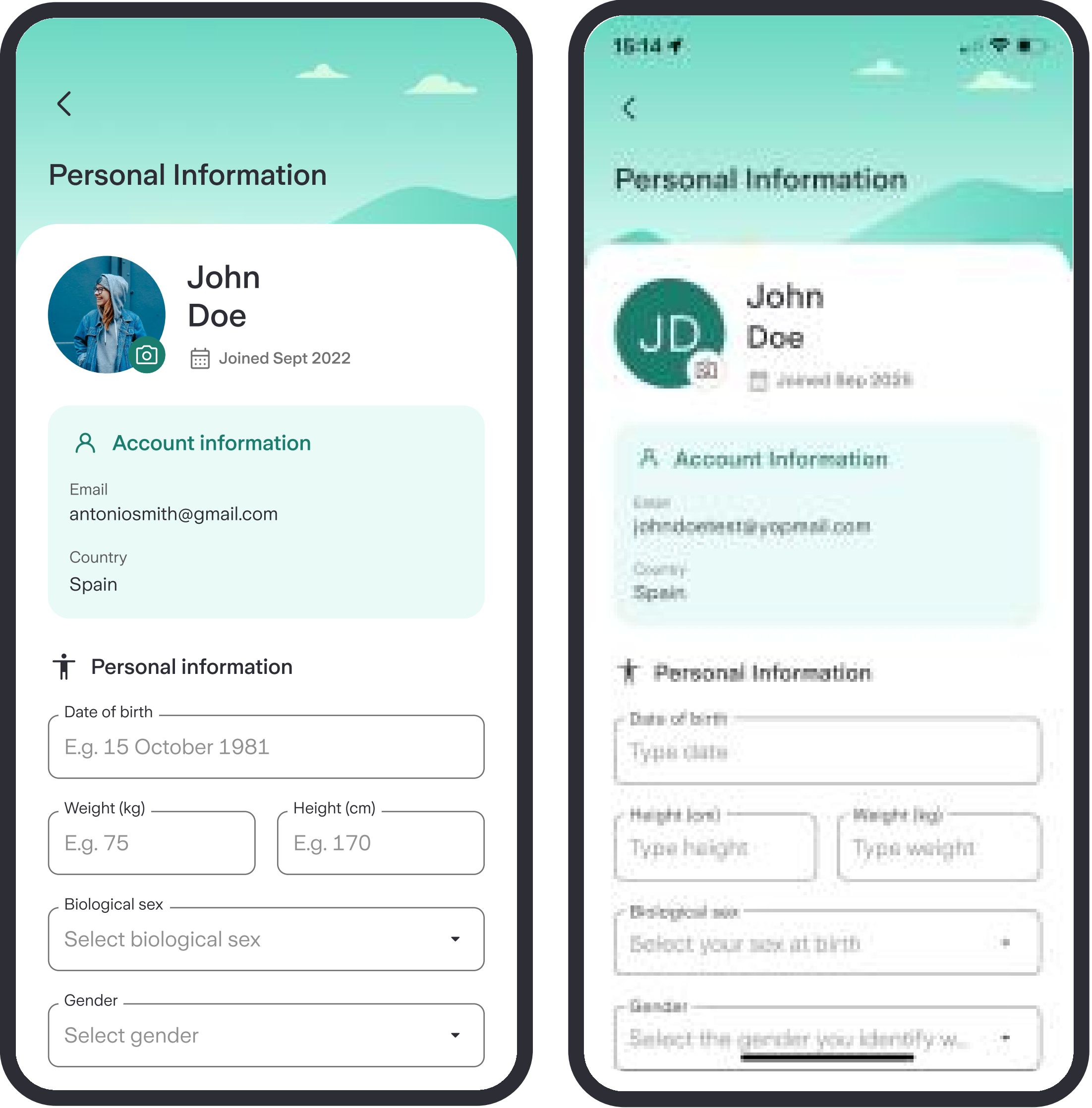
Live Well - Before



Figma

App

Live Well - After



Figma

App

## ✗ Without Design Systems

### **INCONSISTENCY**

Users confused by inconsistent UX

### **INEFFICIENCY**

Recreating & building the same and duplicated components

### **SLOW CHANGE**

Very hard to efficiently propagate changes

### **TECH DEBT**

Time wasted fixing design and technical debt.

### **HIGH COST & DELAY**

Rework and inefficiencies slows down delivery.

**VS.**

## ✓ With Design System

### **CONSISTENCY**

One source of truth, unified, familiar user experience

### **EFFICIENCY**

Build once, use everywhere

### **CHANGE MANAGEMENT**

Easy to propagate changes from design to code

### **QUALITY**

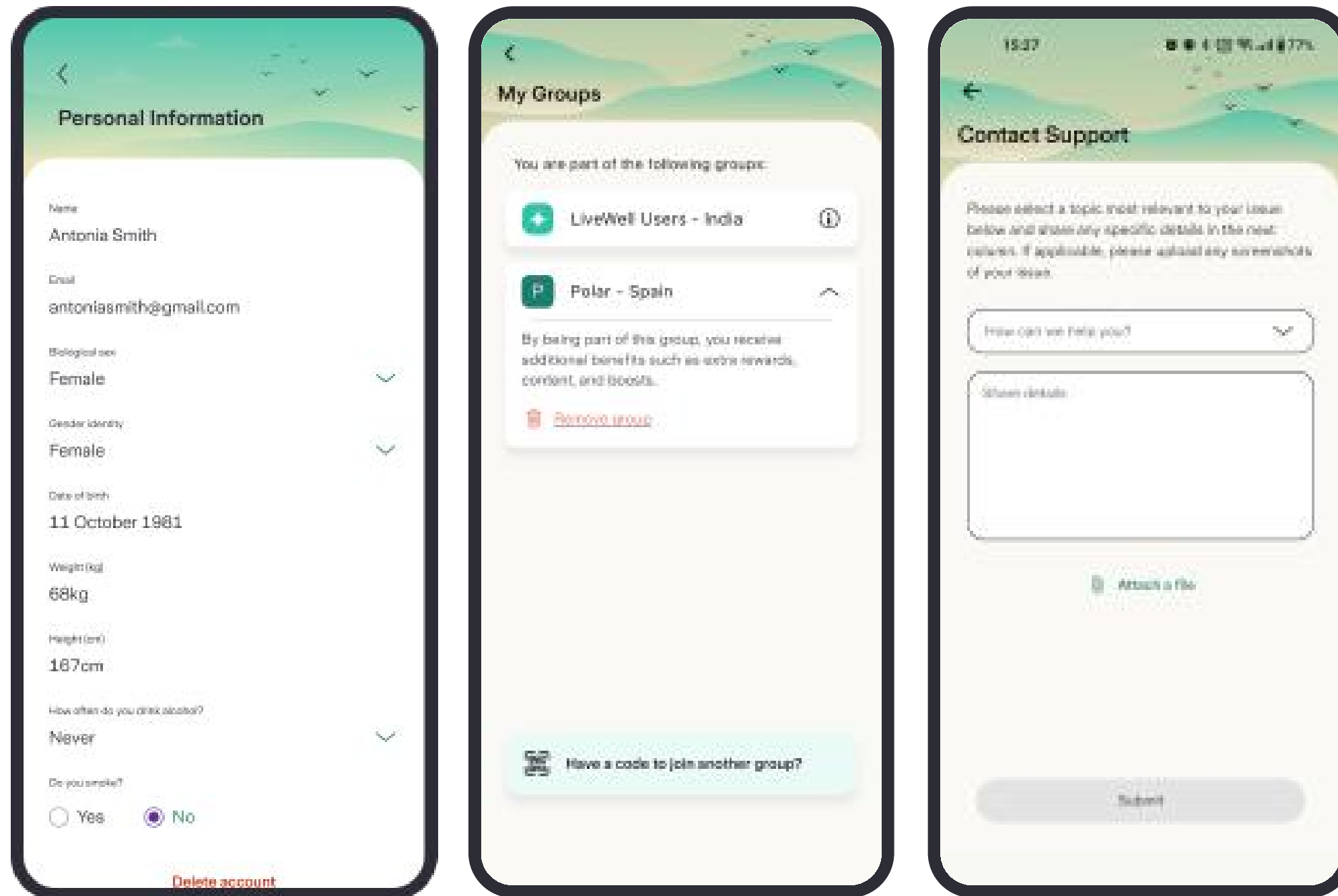
Tested, accessible components out of the box

### **SPEED & FOCUS**

Teams focus on building unique features, not UI details

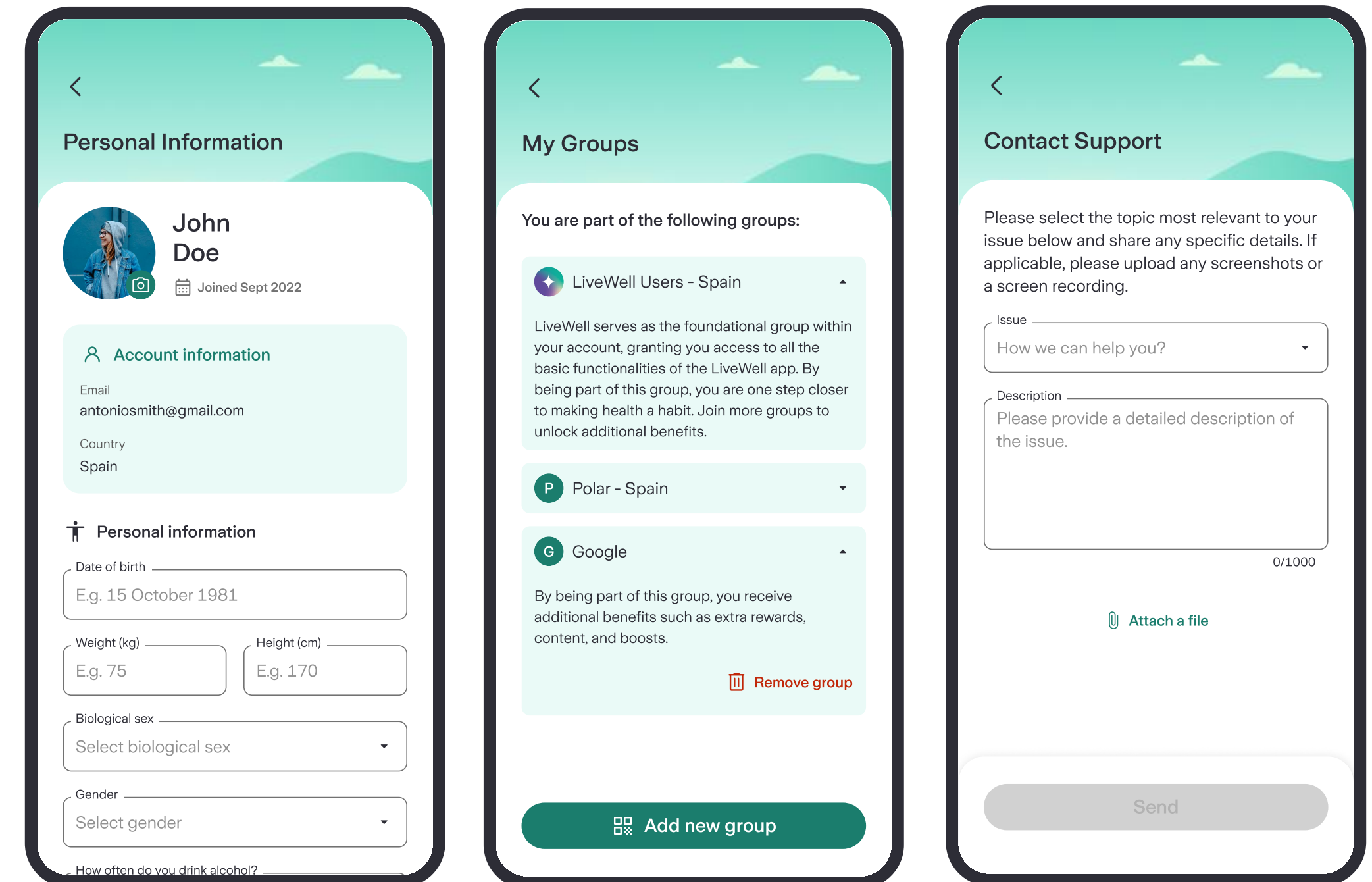


## Before



- Inconsistent UI, duplicate components, different styles, fonts, colors etc.
- No support for dark mode or accessibility.
- Repeated work across teams.

## After



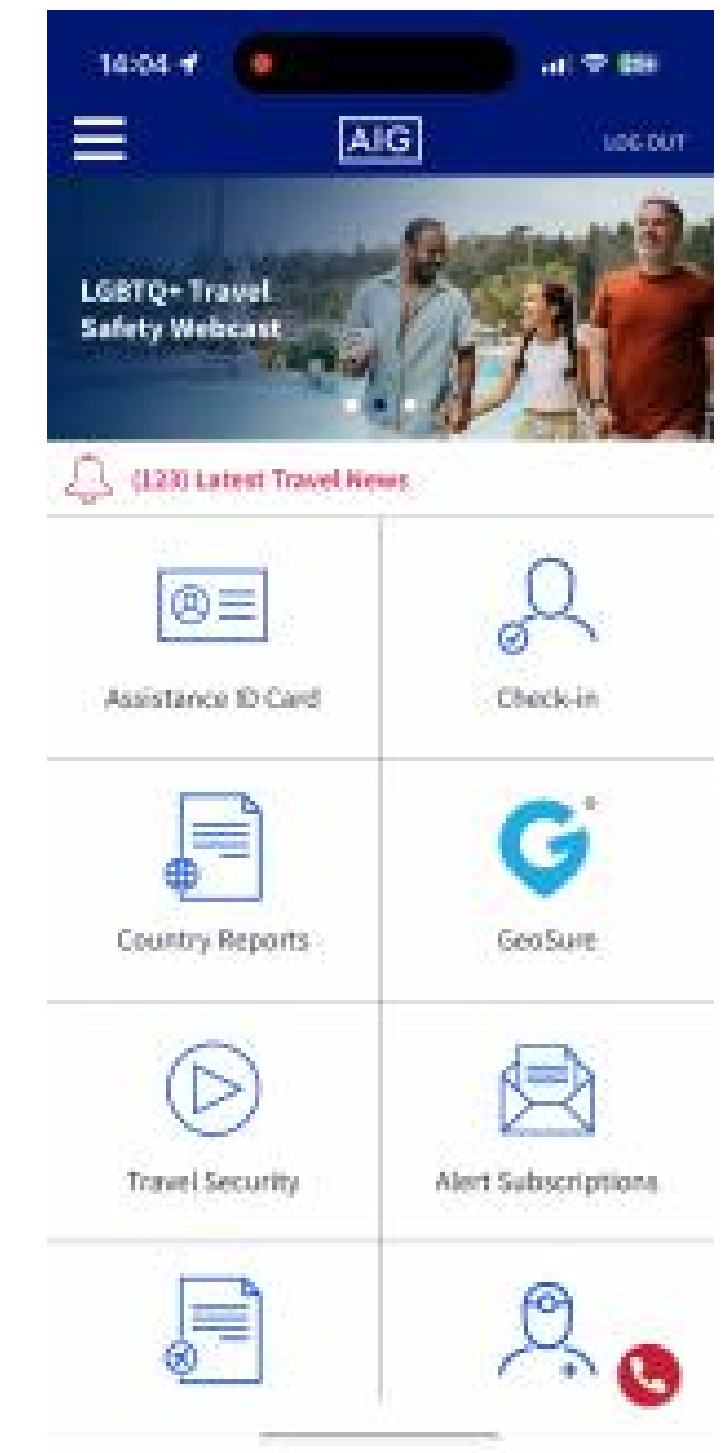
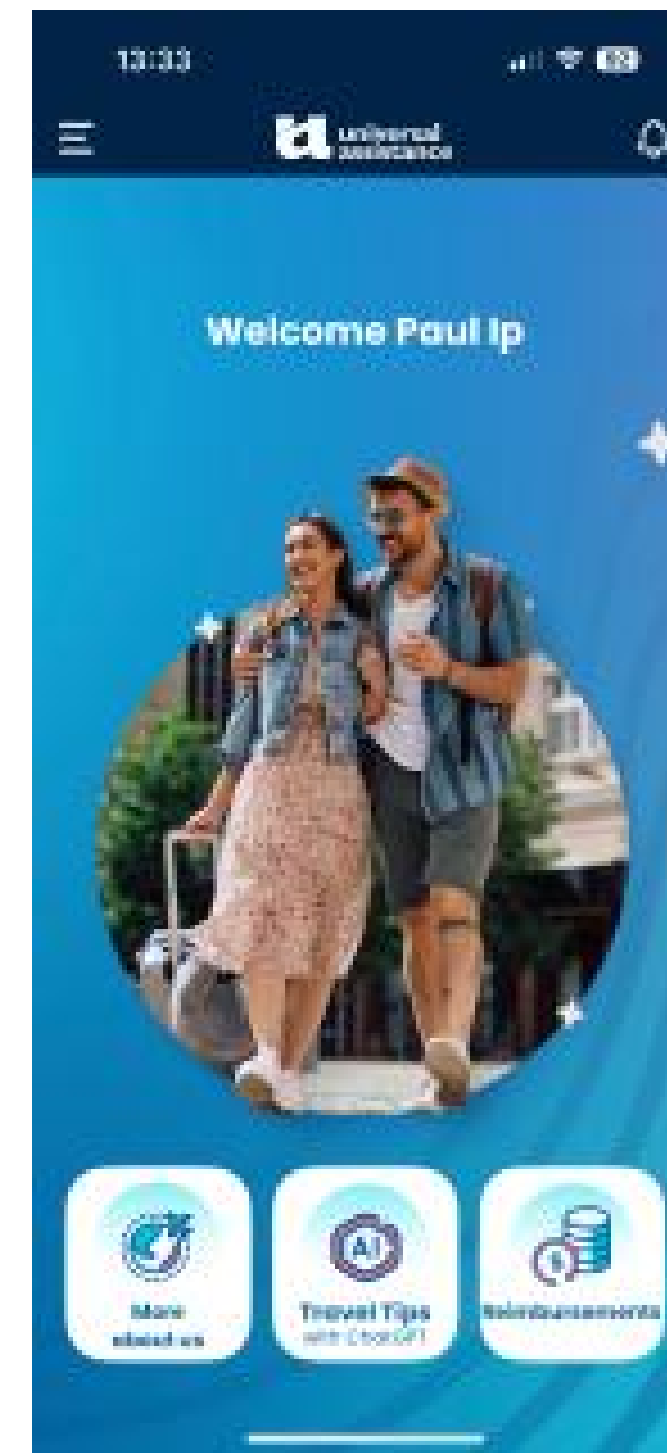
- Unified look with consistent DS components.
- Components accessibility tested and dark mode support by default.
- Reusable components

# Challenges without a shared Design System



## Design POV:

- Inconsistent UI patterns across apps: brand feels fragmented.
- Designers reinvent solutions for each product.
- Inconsistent performance and UX across platforms.
- Risk of poor accessibility standards



## Development POV:

- Duplication of code across apps.
- Higher maintenance → bug fixing in one app doesn't propagate to others.
- Slower delivery → every feature requires reimplementation.
- No behavior or style rules defined. Devs needed to add style, colors etc for each instance.

```
const normalBorder = OutlineInputBorder({
  borderRadius: BorderRadius.all(
    Radius.circular(Dimensions.borderRadiusTextFields),
  ), // BorderRadius.all
  borderSide: BorderSide(
    color: LivewellColors.nearBlack,
  ), // BorderSide
}); // OutlineInputBorder

const errorBorder = OutlineInputBorder(
  borderRadius: BorderRadius.all(
    Radius.circular(Dimensions.borderRadiusTextFields),
  ), // BorderRadius.all
  borderSide: BorderSide(
    color: LivewellColors.peachRed,
  ), // BorderSide
); // OutlineInputBorder

return LVTextFormField(
  labelText: localization.emailFieldLabel,
  controller: emailInputController,
  textInputActions: TextInputAction.next,
  labelSpacing: 8,
  labelTextStyle: theme.labelSmall,
  style: theme.bodyMedium,
  primaryColour: LivewellColors.nearBlack,
  inputValidator: InputValidator(
    (final value) => _validateEmail(
      input: value,
      localization: localization,
    ),
  ), // InputValidator
  decoration: InputDecoration(
    contentPadding:
      const EdgeInsets.symmetric(horizontal: Dimensions.padding16),
    enabledBorder: normalBorder,
    focusedBorder: normalBorder,
    errorBorder: errorBorder,
    focusedErrorBorder: errorBorder,
    hintText: localization.emailExample,
    hintStyle: const LivewellTheme()
      .getRedesignCustomTheme(context: context)
      .textTheme
      .labelSmall,
    errorMaxLines: 2,
  ), // InputDecoration
); // LVTextFormField
```

LW component without Design system

```
LDSInputSmall(
  labelText: 'Email',
  hintText: 'Your email',
  controller: textEditingControllerEmail,
  autofillHints: const <String>[AutofillHints.email],
  validator: _validateEmail,
  keyboardType: TextInputType.emailAddress,
  textInputAction: TextInputAction.next,
  onEditingComplete: passwordFocus.requestFocus,
  maxLength: 30,
  focusNode: emailFocus,
), // LDSInputSmall
```

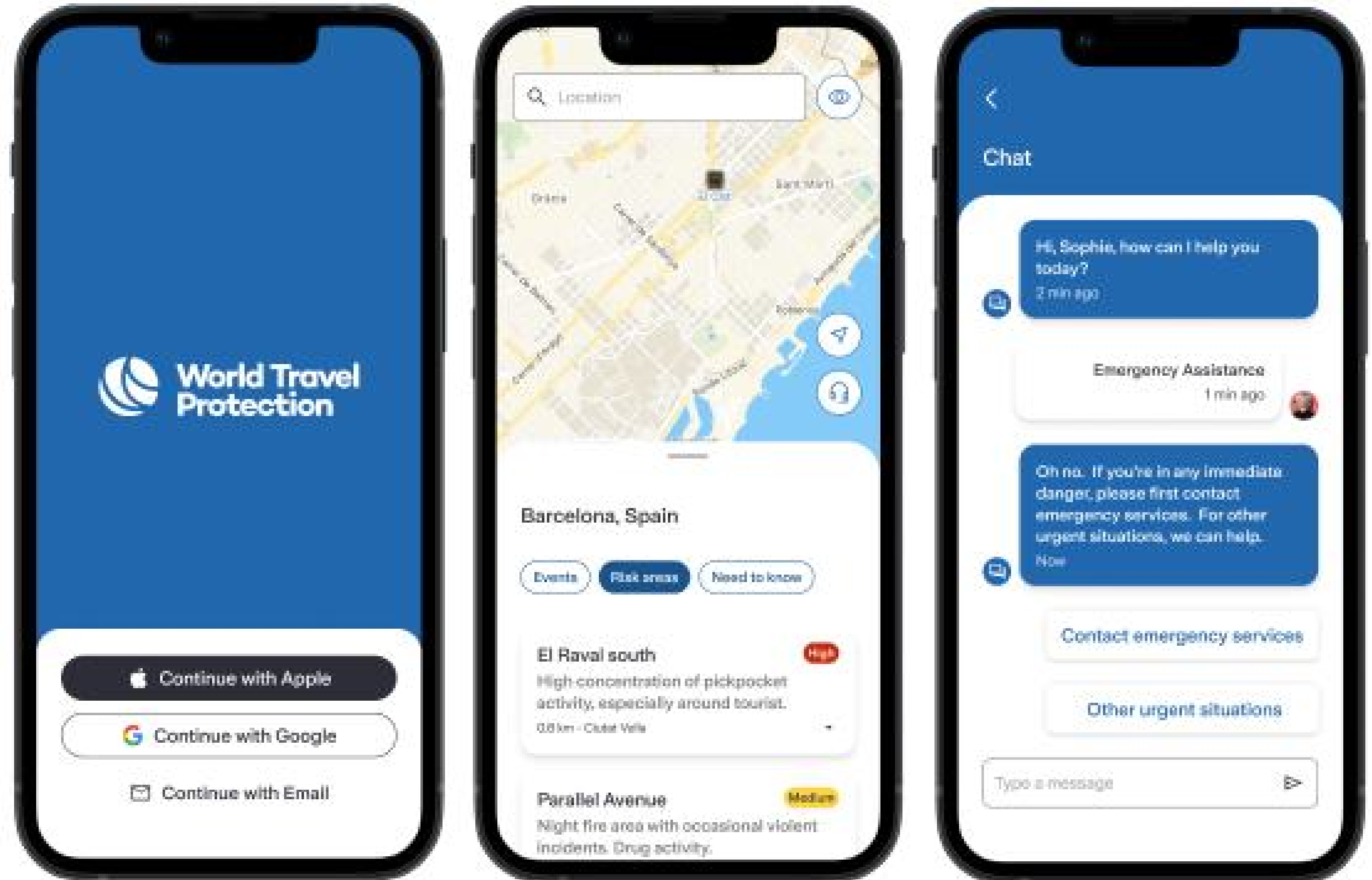
LW component with Design system

Clear example of what can we do:



# WTP Hackathon

A clear example of what's possible using a DS it's our **speed** and **efficiency** in our previous work with the **World Travel Protection Hackathon**



# Flutter & Design Systems

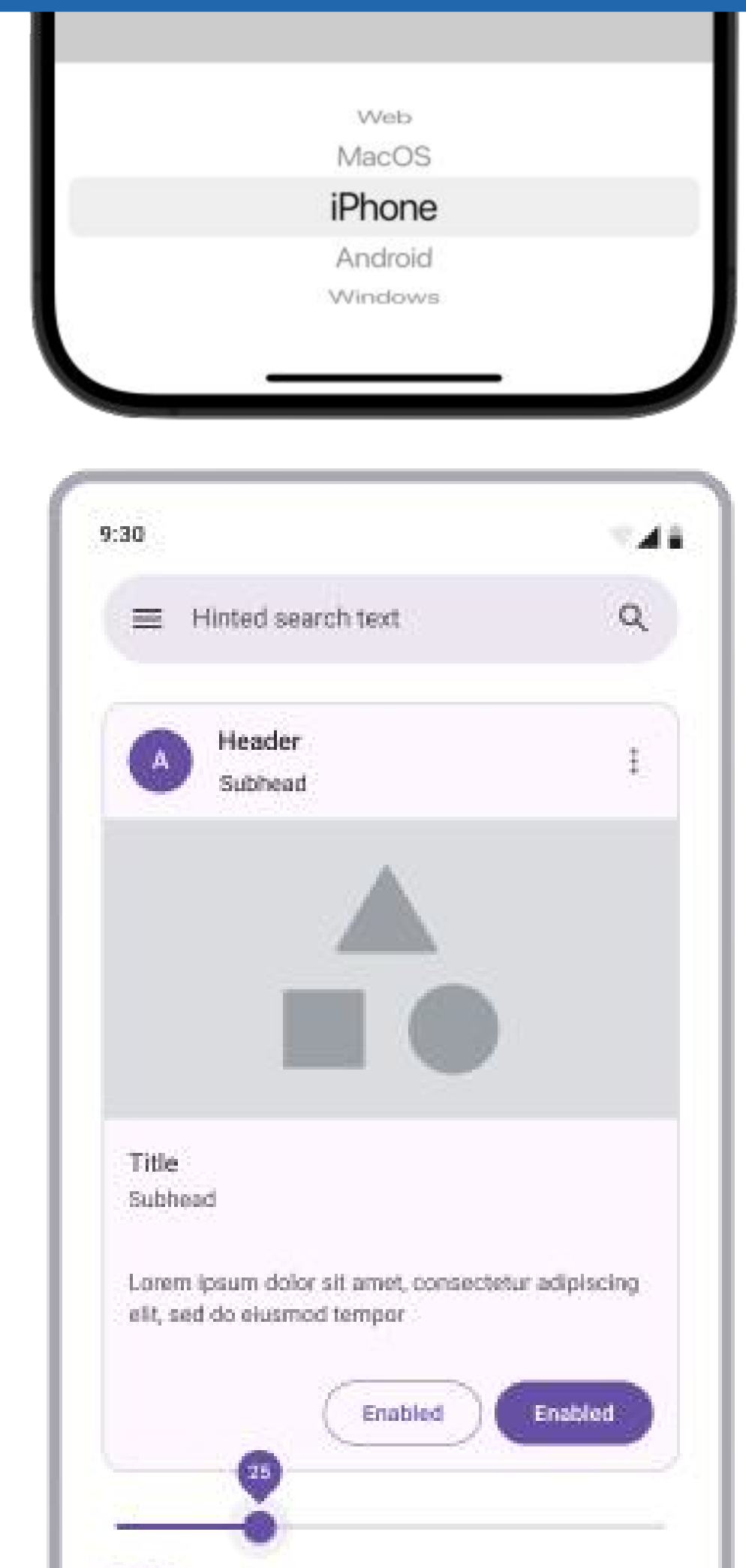
# Flutter /Material /Cupertino

“

Flutter is Google’s UI framework for building apps on iOS, Android, Web, and Desktop - all from a single codebase.

It comes with built-in widgets that follow two popular Design Systems (For an native look on both platforms):

- **Material Design** - Google’s design language, default on Android
- **Cupertino Design** - Apple’s design language (Flutter’s implementation), default on iOS



Material



Cupertino

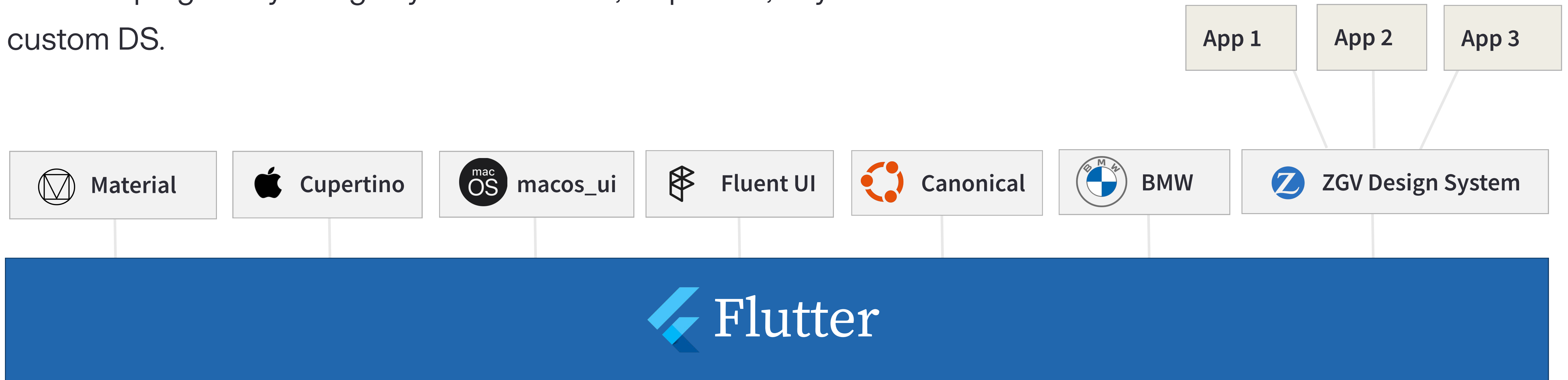


# Decoupling Material & Cupertino

**New approach:** Decouple design systems from the core framework.

Flutter becomes a neutral framework (not just “Material-first”).

Easier to plug in any design system: Material, Cupertino, or your brand’s custom DS.



Why a multi-app design system?

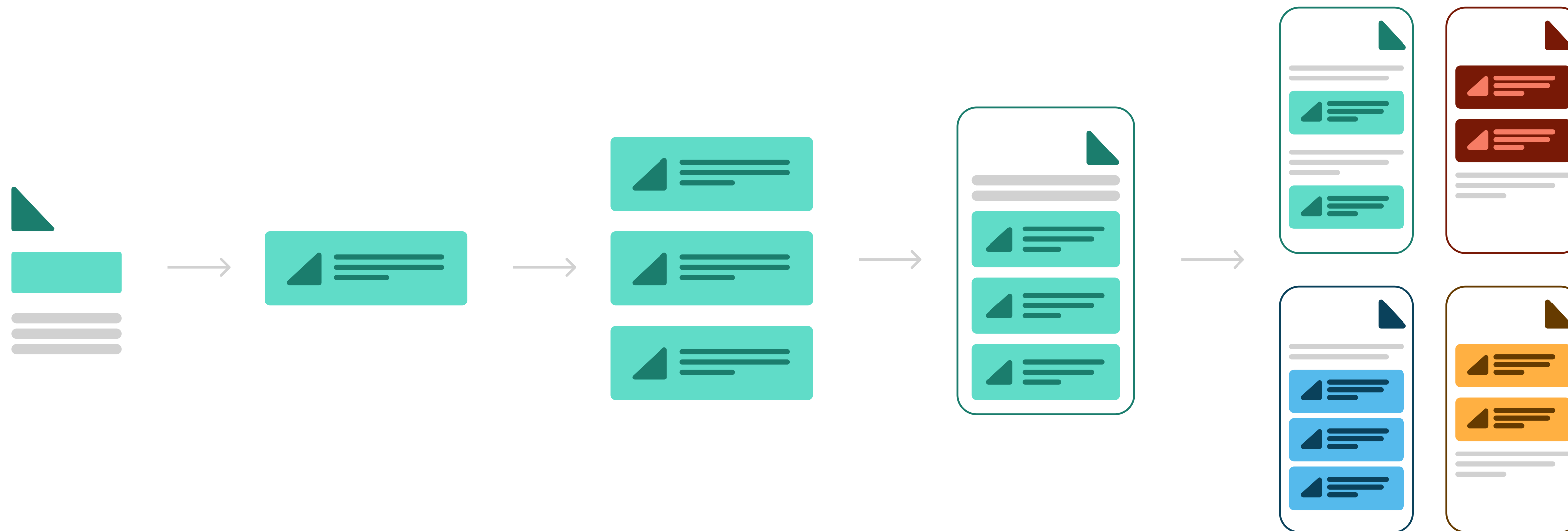
# Reusability loves modularity

## Benefits for Multi-App Ecosystem:

Core elements will compose complex components and the whole App experience will be more consistent.

## One DS → Multiple Apps

- Shared brand identity across all products.
- Faster launch of new apps using the same DS.
- Easier maintenance: update DS once → all apps benefit.
- Consistent UX even with different teams/projects.
- Scales design and dev resources efficiently.





# Conclusion

- A **unified** Design System bridging **design and development**.
- Transformed LiveWell **from inconsistent UI to cohesive** experience.
- Consistency: **One source** of truth **across all apps**.
- Enabled **dark mode, accessibility, and multi-platform** consistency by default.
- Created foundation for **quality multi-app development for ZGV apps**.

Thank you!